2.19 Hyperbolic Functions: Differentiation

Worked Solutions

1 a)

$$\begin{aligned}\frac{d}{dx}(\sinh x) &= \frac{d}{dx}\left(\frac{e^x - e^{-x}}{2}\right) \\ &= \frac{e^x + e^{-x}}{2} = \cosh x.\end{aligned}$$

1 b)

$$\begin{aligned}\frac{d}{dx}(\cosh x) &= \frac{d}{dx}\left(\frac{e^x + e^{-x}}{2}\right) \\ &= \frac{e^x - e^{-x}}{2} = \sinh x.\end{aligned}$$

1 c)

$$\begin{aligned}\frac{d}{dx}(\tanh x) &= \frac{d}{dx}\left(\frac{e^{2x} - 1}{e^{2x} + 1}\right) \\ &= \frac{(2e^{2x})(e^{2x} + 1) - (e^{2x} - 1)(2e^{2x})}{(e^{2x} + 1)^2} \\ &= \frac{4e^{2x}}{e^{4x} + 2e^{2x} + 1} = \frac{4}{e^{2x} + 2 + e^{-2x}} \\ &= \frac{4}{(e^x + e^{-x})^2} = \left(\frac{2}{e^x + e^{-x}}\right)^2 \\ &= \left(\frac{1}{\cosh x}\right)^2 = \operatorname{sech}^2 x.\end{aligned}$$

2 a)

Let $f(x) = \sinh x$, so $f(x + h) = \sinh(x + h)$. By first principles,

$$\begin{aligned}f'(x) &= \lim_{h \to 0} \frac{f(x + h) - f(x)}{h} \\ &= \lim_{h \to 0} \frac{\sinh(x + h) - \sinh x}{h} \\ &= \lim_{h \to 0} \frac{\sinh x \cosh h + \cosh x \sinh h - \sinh x}{h} \\ &= \sinh x \left(\lim_{h \to 0} \frac{\cosh h - 1}{h}\right) + \cosh x \left(\lim_{h \to 0} \frac{\sinh h}{h}\right) \\ &= \sinh x \times 0 + \cosh x \times 1 = \cosh x.\end{aligned}$$

2 b)

Let $f(x) = \cosh x$, so $f(x+h) = \cosh(x+h)$. By first principles,

$$\begin{aligned} f'(x) &= \lim_{h\to 0} \frac{\cosh(x+h) - \cosh x}{h} \\ &= \lim_{h\to 0} \frac{\cosh x \cosh h + \sinh x \sinh h - \cosh x}{h} \\ &= \cosh x \left(\lim_{h\to 0} \frac{\cosh h - 1}{h}\right) + \sinh x \left(\lim_{h\to 0} \frac{\sinh h}{h}\right) \\ &= \cosh x \times 0 + \sinh x \times 1 = \sinh x. \end{aligned}$$

3 a)

$y = \sinh 2x$, so $\dfrac{dy}{dx} = 2\cosh 2x$.

3 b)

$y = \cosh 3x$, so $\dfrac{dy}{dx} = 3\sinh 3x$.

3 c)

$y = \tanh 6x$, so $\dfrac{dy}{dx} = 6\,\mathrm{sech}^2\, 6x$.

3 d)

$y = \sinh \frac{1}{4}x$, so $\dfrac{dy}{dx} = \frac{1}{4}\cosh \frac{1}{4}x$.

3 e)

$y = \cosh \frac{x}{3}$, so $\dfrac{dy}{dx} = \frac{1}{3}\sinh \frac{x}{3}$.

3 f)

$y = \tanh\left(-\frac{x}{5}\right)$, so $\dfrac{dy}{dx} = -\frac{1}{5}\,\mathrm{sech}^2\left(-\frac{x}{5}\right)$. Alternatively, $y = -\tanh \frac{x}{5}$ and $\dfrac{dy}{dx} = -\frac{1}{5}\,\mathrm{sech}^2 \frac{x}{5}$.

4 a)

$y = 3\sinh 4x - 6\tanh 3x$, so $\dfrac{dy}{dx} = 12\cosh 4x - 18\,\mathrm{sech}^2\, 3x$.

4 b)

$y = 5\cosh \frac{x}{10} + \frac{1}{2}\tanh \frac{x}{4}$, so $\dfrac{dy}{dx} = \frac{1}{2}\sinh \frac{x}{10} + \frac{1}{8}\,\mathrm{sech}^2 \frac{x}{4}$.

5)

$$\begin{aligned} y &= 3\tanh \frac{1}{2}x, \\ \frac{dy}{dx} &= \frac{3}{2}\,\mathrm{sech}^2 \frac{1}{2}x, \end{aligned}$$

$$\begin{aligned}\left.\frac{dy}{dx}\right|_{x=\ln 16} &= \frac{3}{2}\operatorname{sech}^2\left(\frac{1}{2}\ln 16\right) = \frac{3}{2}\operatorname{sech}^2(\ln 4)\\ &= \frac{3}{2}\frac{1}{(\cosh(\ln 4))^2} = \frac{3}{2}\frac{1}{\left(\frac{e^{\ln 4}+e^{-\ln 4}}{2}\right)^2}\\ &= \frac{3}{2}\frac{1}{\left(\frac{4+\frac{1}{4}}{2}\right)^2} = \frac{96}{289}.\end{aligned}$$

6)

$$\begin{aligned}y &= 3\sinh x,\\ y(\ln 6) &= \frac{3}{2}\left(6-\frac{1}{6}\right) = \frac{35}{4},\\ \frac{dy}{dx} &= 3\cosh x,\\ \left.\frac{dy}{dx}\right|_{x=\ln 6} &= \frac{3}{2}\left(6+\frac{1}{6}\right) = \frac{37}{4}.\end{aligned}$$

The tangent is $y-\frac{35}{4}=\frac{37}{4}(x-\ln 6)$, so $-37x+4y=35-37\ln 6$ or $37x-4y=37\ln 6-35$.

7)

$$\begin{aligned}y(\ln 2) &= 5\sinh(\ln 2)-\cosh(\ln 2)\\ &= \frac{5}{2}\left(2-\frac{1}{2}\right)-\frac{1}{2}\left(2+\frac{1}{2}\right) = \frac{5}{2},\\ \frac{dy}{dx} &= 5\cosh x-\sinh x,\\ \left.\frac{dy}{dx}\right|_{x=\ln 2} &= \frac{5}{2}\left(2+\frac{1}{2}\right)-\frac{1}{2}\left(2-\frac{1}{2}\right) = \frac{11}{2}.\end{aligned}$$

The tangent is $y-\frac{5}{2}=\frac{11}{2}(x-\ln 2)$, so $y=\frac{11}{2}x+\frac{5}{2}-\frac{11}{2}\ln 2$.

8)

$$\begin{aligned}y &= 3\cosh 2x-4\sinh 3x,\\ y\big|_{x=\ln(1/3)} &= \frac{3}{2}\left(\frac{1}{9}+9\right)-2\left(\frac{1}{27}-27\right) = \frac{1825}{27},\\ \frac{dy}{dx} &= 6\sinh 2x-12\cosh 3x,\\ \left.\frac{dy}{dx}\right|_{x=\ln(1/3)} &= 3\left(\frac{1}{9}-9\right)-6\left(\frac{1}{27}+27\right) = -\frac{1700}{9}.\end{aligned}$$

The tangent is $y-\frac{1825}{27}=-\frac{1700}{9}(x-\ln\frac{1}{3})$, hence $y=-\frac{1700}{9}x+\frac{1825}{27}+\frac{1700}{9}\ln\frac{1}{3}$.

9)

$$y(\ln 4) = \tanh\left(\frac{1}{2}\ln 4\right) = \frac{2-\frac{1}{2}}{2+\frac{1}{2}} = \frac{3}{5},$$

$$\frac{dy}{dx} = \frac{1}{2}\operatorname{sech}^2\frac{x}{2},$$
$$\left.\frac{dy}{dx}\right|_{x=\ln 4} = \frac{1}{2}\left(\frac{2}{2+\frac{1}{2}}\right)^2 = \frac{8}{25}.$$

The tangent is $y - \frac{3}{5} = \frac{8}{25}(x - \ln 4)$, so $y = \frac{8}{25}x + \frac{3}{5} - \frac{8}{25}\ln 4$.

10)

$$y(0) = 8 + 6\tanh 0 - 4\tanh 0 = 8,$$
$$\frac{dy}{dx} = 2\operatorname{sech}^2\frac{x}{3} - \frac{8}{3}\operatorname{sech}^2\frac{2x}{3},$$
$$\left.\frac{dy}{dx}\right|_{x=0} = 2 - \frac{8}{3} = -\frac{2}{3}.$$

The normal gradient is $\frac{3}{2}$, so the normal is $y - 8 = \frac{3}{2}(x - 0)$, or $y = \frac{3}{2}x + 8$.

11)

For l_1, $y = 12\sinh\frac{x}{3}$ and at $x = \ln 8 = 3\ln 2$, $y = 12\sinh(\ln 2) = 9$. Its gradient is $4\cosh\frac{x}{3}$, which is 5 there. Thus the normal gradient is $-\frac{1}{5}$ and

$$l_1: \quad y - 9 = -\frac{1}{5}(x - \ln 8), \qquad y = -\frac{1}{5}x + \frac{1}{5}\ln 8 + 9.$$

For l_2, $y = 6\cosh\frac{x}{4}$ and at $x = \ln 16 = 4\ln 2$, $y = 6\cosh(\ln 2) = \frac{15}{2}$. Its gradient is $\frac{3}{2}\sinh\frac{x}{4}$, which is $\frac{3}{4}$ there. Thus the normal gradient is $-\frac{4}{3}$ and

$$l_2: \quad y - \frac{15}{2} = -\frac{4}{3}(x - \ln 16), \qquad y = -\frac{8}{9}x + \frac{8}{9}\ln 16 + \frac{15}{2}.$$

At the intersection,

$$\begin{aligned} -\frac{1}{5}x + \frac{1}{5}\ln 8 + 9 &= -\frac{8}{9}x + \frac{8}{9}\ln 16 + \frac{15}{2} \\ \frac{31}{45}x &= \frac{8}{9}\ln 16 - \frac{1}{5}\ln 8 - \frac{3}{2} \\ &= \frac{32}{9}\ln 2 - \frac{3}{5}\ln 2 - \frac{3}{2} \\ &= \frac{133}{45}\ln 2 - \frac{3}{2}, \\ x &= \frac{133}{31}\ln 2 - \frac{135}{62}. \end{aligned}$$

12)

$$y = \sinh x + \tanh x,$$
$$\frac{dy}{dx} = \cosh x + \operatorname{sech}^2 x.$$

For all real x, $\cosh x \geq 1$ and $0 < \operatorname{sech}^2 x \leq 1$, so $\cosh x + \operatorname{sech}^2 x > 1$. Hence $\frac{dy}{dx} > 0$ for all x, and the curve is always increasing.

13)

$$\begin{aligned} y &= \sinh x + 2\cosh x, \\ \frac{dy}{dx} &= \cosh x + 2\sinh x. \end{aligned}$$

At a stationary point, $\cosh x + 2\sinh x = 0$, so $1 + 2\tanh x = 0$, $\tanh x = -\frac{1}{2}$, and $x = \frac{1}{2}\ln\frac{1}{3} = \ln\frac{1}{\sqrt{3}}$.

$$\begin{aligned} y\big|_{x=\ln(1/\sqrt{3})} &= \sinh\left(\ln\frac{1}{\sqrt{3}}\right) + 2\cosh\left(\ln\frac{1}{\sqrt{3}}\right) \\ &= \frac{1/\sqrt{3} - \sqrt{3}}{2} + \left(\frac{1}{\sqrt{3}} + \sqrt{3}\right) = \sqrt{3}. \end{aligned}$$

The stationary point is $\left(\ln\frac{1}{\sqrt{3}}, \sqrt{3}\right)$.

14)

$$\begin{aligned} y &= 9\cosh x + 3\sinh 2x, \\ \frac{dy}{dx} &= 9\sinh x + 6\cosh 2x \\ &= 9\sinh x + 6(1 + 2\sinh^2 x) = 12\sinh^2 x + 9\sinh x + 6. \end{aligned}$$

If the gradient is 1, then $12\sinh^2 x + 9\sinh x + 5 = 0$. Its discriminant is $9^2 - 4(12)(5) = -159 < 0$, so there is no real value of $\sinh x$ satisfying this equation, and no point on the curve where the gradient is 1.

15)

$$\begin{aligned} y &= p\sinh x - q\cosh x, \\ \frac{dy}{dx} &= p\cosh x - q\sinh x. \end{aligned}$$

At a stationary point, $p\cosh x - q\sinh x = 0$, so $p - q\tanh x = 0$ and $\tanh x = \frac{p}{q}$. If $p \geq q$, then $\frac{p}{q} \geq 1$. But $-1 < \tanh x < 1$, so there are no real solutions; the curve has no stationary points.

16)

$$\begin{aligned} y &= \sinh x - 8\tanh x, \\ \frac{dy}{dx} &= \cosh x - 8\,\mathrm{sech}^2 x. \end{aligned}$$

At a stationary point,

$$\begin{aligned} \cosh x - \frac{8}{\cosh^2 x} &= 0 \\ \Rightarrow \cosh^3 x &= 8 \\ \Rightarrow \cosh x &= 2 \\ \Rightarrow x &= \pm\ln(2 + \sqrt{3}) = \ln(2 \pm \sqrt{3}). \end{aligned}$$

For $x = \ln(2+\sqrt{3})$,

$$\begin{aligned} y &= \sinh(\ln(2+\sqrt{3})) - 8\tanh(\ln(2+\sqrt{3})) \\ &= \frac{(2+\sqrt{3})-(2-\sqrt{3})}{2} - 8\frac{(2+\sqrt{3})-(2-\sqrt{3})}{(2+\sqrt{3})+(2-\sqrt{3})} \\ &= \sqrt{3} - 4\sqrt{3} = -3\sqrt{3}. \end{aligned}$$

For $x = \ln(2-\sqrt{3})$,

$$\begin{aligned} y &= \frac{(2-\sqrt{3})-(2+\sqrt{3})}{2} - 8\frac{(2-\sqrt{3})-(2+\sqrt{3})}{(2-\sqrt{3})+(2+\sqrt{3})} \\ &= -\sqrt{3} + 4\sqrt{3} = 3\sqrt{3}. \end{aligned}$$

Thus the stationary points are $(\ln(2+\sqrt{3}), -3\sqrt{3})$ and $(\ln(2-\sqrt{3}), 3\sqrt{3})$.

17)

$$\begin{aligned} y &= 5\cosh x + \sinh x + x, \\ \frac{dy}{dx} &= 5\sinh x + \cosh x + 1. \end{aligned}$$

At a stationary point,

$$\begin{aligned} 5\sinh x + \cosh x + 1 &= 0 \\ \Rightarrow 3e^{x} - 2e^{-x} + 1 &= 0 \\ \Rightarrow 3e^{2x} + e^{x} - 2 &= 0 \\ \Rightarrow (3e^{x} - 2)(e^{x} + 1) &= 0. \end{aligned}$$

$e^x = -1$ has no real solutions. Therefore the only possibility is $e^x = \frac{2}{3}$, so there is a single stationary point when $x = \ln\frac{2}{3}$.

18)

$$\begin{aligned} y &= \cosh 3x - 2\sinh x, \\ \frac{d^2y}{dx^2} &= 9\cosh 3x - 2\sinh x. \end{aligned}$$

At $x = \ln 2$,

$$\begin{aligned} \left.\frac{d^2y}{dx^2}\right|_{x=\ln 2} &= 9\cosh(\ln 8) - 2\sinh(\ln 2) \\ &= 9\frac{8+1/8}{2} - 2\frac{2-1/2}{2} = \frac{585}{16} - \frac{3}{2} = \frac{561}{16} > 0. \end{aligned}$$

The curve is convex (concave upwards) at $x = \ln 2$.

19)

$$y = 4\sinh 2x - 3\cosh 2x,$$

$$\frac{dy}{dx} = 8\cosh 2x - 6\sinh 2x,$$
$$\frac{d^2y}{dx^2} = 16\sinh 2x - 12\cosh 2x.$$

Points of inflection satisfy $16\sinh 2x - 12\cosh 2x = 0$, so $\tanh 2x = \frac{3}{4}$ and

$$x = \frac{1}{4}\ln\left(\frac{1+\frac{3}{4}}{1-\frac{3}{4}}\right) = \frac{1}{4}\ln 7.$$

$$\begin{aligned} y\big|_{x=\frac{1}{4}\ln 7} &= 4\sinh\left(\frac{1}{2}\ln 7\right) - 3\cosh\left(\frac{1}{2}\ln 7\right) \\ &= 2\left(\sqrt{7} - \frac{1}{\sqrt{7}}\right) - \frac{3}{2}\left(\sqrt{7} + \frac{1}{\sqrt{7}}\right) = 0. \end{aligned}$$

At $x = 0$, $\frac{d^2y}{dx^2} = -12 < 0$, while at $x = 1$, $\frac{d^2y}{dx^2} = 16\sinh 1 - 12\cosh 1 = 0.286\ldots > 0$. Thus $\left(\frac{1}{4}\ln 7, 0\right)$ is a point of inflection.

20 a)

$$y = \sinh(\sqrt{x}) = \sinh(x^{1/2}),$$
$$\frac{dy}{dx} = \frac{1}{2}x^{-1/2}\cosh(x^{1/2}).$$

20 b)

$$y = \cosh\left(\frac{1}{\sqrt{x}}\right) = \cosh(x^{-1/2}),$$
$$\frac{dy}{dx} = -\frac{1}{2}x^{-3/2}\sinh(x^{-1/2}).$$

20 c)

$$y = \tanh\left(\frac{x+1}{2\sqrt{x}}\right) = \tanh\left(\frac{1}{2}x^{1/2} + \frac{1}{2}x^{-1/2}\right),$$
$$\frac{dy}{dx} = \left(\frac{1}{4}x^{-1/2} - \frac{1}{4}x^{-3/2}\right)\operatorname{sech}^2\left(\frac{x+1}{2\sqrt{x}}\right).$$

21 a)

$y = \sinh^2 x = (\sinh x)^2$, so $\dfrac{dy}{dx} = 2\sinh x\cosh x$.

21 b)

$y = \cosh^2 x = (\cosh x)^2$, so $\dfrac{dy}{dx} = 2\sinh x\cosh x$.

21 c)

$y = \sinh^3 x = (\sinh x)^3$, so $\dfrac{dy}{dx} = 3\sinh^2 x \cosh x$.

21 d)

$y = \cosh^4 x = (\cosh x)^4$, so $\dfrac{dy}{dx} = 4\sinh x \cosh^3 x$.

21 e)

$y = \tanh^2 x = (\tanh x)^2$, so $\dfrac{dy}{dx} = 2\operatorname{sech}^2 x \tanh x$.

21 f)

$y = \tanh^5 x = (\tanh x)^5$, so $\dfrac{dy}{dx} = 5\operatorname{sech}^2 x \tanh^4 x$.

22)

$$\begin{aligned} y &= 3\cosh^2 x - \sinh x, \\ \frac{dy}{dx} &= 6\sinh x \cosh x - \cosh x = \cosh x(6\sinh x - 1). \end{aligned}$$

Since $\cosh x = 0$ has no real solutions, $6\sinh x - 1 = 0$, so $\sinh x = \frac{1}{6}$ and

$$x = \ln\left(\frac{1}{6} + \sqrt{\left(\frac{1}{6}\right)^2 + 1}\right) = \ln\left(\frac{1+\sqrt{37}}{6}\right).$$

There is a single stationary point at this x-coordinate.

23)

$$\begin{aligned} f(x) &= \cosh^2 2x - p\cosh 2x, \\ f'(x) &= 4\sinh 2x \cosh 2x - 2p\sinh 2x \\ &= 2\sinh 2x(2\cosh 2x - p). \end{aligned}$$

The first factor is zero at $x = 0$ for all real p. The second factor has real roots if $p \geq 2$, since $\cosh 2x \geq 1$. Therefore, the minimum value of p for the curve to have more than one stationary point is $p = 2$.

24 a)

$y = \sinh(\cosh x)$, so $\dfrac{dy}{dx} = \sinh x \cosh(\cosh x)$.

24 b)

$y = \cosh(\sinh x)$, so $\dfrac{dy}{dx} = \cosh x \sinh(\sinh x)$.

24 c)

$y = \tanh(\tanh x)$, so $\dfrac{dy}{dx} = \operatorname{sech}^2 x \operatorname{sech}^2(\tanh x)$.

25 a)

$$\frac{d}{dx}(\ln(\sinh x)) = \frac{\cosh x}{\sinh x} = \coth x.$$

25 b)

$$\frac{d}{dx}(\ln(\cosh x)) = \frac{\sinh x}{\cosh x} = \tanh x.$$

26)

$$\begin{aligned}\frac{d}{dx}\left(\frac{1}{2}\ln(\tanh x)\right) &= \frac{1}{2}\frac{\operatorname{sech}^2 x}{\tanh x} \\ &= \frac{1}{2}\frac{1}{\cosh^2 x}\frac{\cosh x}{\sinh x} \\ &= \frac{1}{2\sinh x \cosh x} = \frac{1}{\sinh 2x} = \operatorname{cosech} 2x.\end{aligned}$$

27 a)

$$\begin{aligned}y &= \sinh x \cosh x, \\ \frac{dy}{dx} &= \sinh^2 x + \cosh^2 x.\end{aligned}$$

Alternatively, $y = \frac{1}{2}\sinh 2x$, so $\frac{dy}{dx} = \cosh 2x$.

27 b)

$y = \sinh 3x \cosh 4x$, so

$$\frac{dy}{dx} = 3\cosh 3x \cosh 4x + 4\sinh 3x \sinh 4x.$$

27 c)

$y = x\sinh x$, so $\frac{dy}{dx} = x\cosh x + \sinh x$.

27 d)

$y = x^2 \cosh x$, so $\frac{dy}{dx} = x^2 \sinh x + 2x\cosh x$.

27 e)

$y = (x+1)\sinh 2x$, so $\frac{dy}{dx} = 2(x+1)\cosh 2x + \sinh 2x$.

27 f)

$y = (1 - x^2)\cosh\frac{x}{2}$, so

$$\frac{dy}{dx} = \frac{1}{2}(1 - x^2)\sinh\frac{x}{2} - 2x\cosh\frac{x}{2}.$$

28)

$$\begin{aligned} y &= \sinh 2x \tanh 3x, \\ \frac{dy}{dx} &= 3\sinh 2x \operatorname{sech}^2 3x + 2\cosh 2x \tanh 3x. \end{aligned}$$

At $x = \ln 2$, this is

$$\begin{aligned} &3\sinh(\ln 4)\operatorname{sech}^2(\ln 8) + 2\cosh(\ln 4)\tanh(\ln 8) \\ &= 3\left(\frac{4 - \frac{1}{4}}{2}\right)\left(\frac{2}{8 + \frac{1}{8}}\right)^2 + 2\left(\frac{4 + \frac{1}{4}}{2}\right)\frac{8 - \frac{1}{8}}{8 + \frac{1}{8}} \\ &= \frac{288}{845} + \frac{1071}{260} = \frac{3015}{676}. \end{aligned}$$

29)

$$\begin{aligned} y(0) &= (0^2 + 0 - 6)\cosh 0 = -6, \\ \frac{dy}{dx} &= (x^2 + x - 6)\sinh x + (2x + 1)\cosh x, \\ \left.\frac{dy}{dx}\right|_{x=0} &= 1. \end{aligned}$$

The tangent is $y + 6 = x$, so $y = x - 6$.

30)

$$\begin{aligned} y(0) &= \sinh(-\ln 2)\cosh(-\ln 3) = -\frac{5}{4}, \\ \frac{dy}{dx} &= 3\sinh(12x - \ln 2)\sinh(3x - \ln 3) + 12\cosh(12x - \ln 2)\cosh(3x - \ln 3), \\ \left.\frac{dy}{dx}\right|_{x=0} &= 3\left(-\frac{3}{4}\right)\left(-\frac{4}{3}\right) + 12\left(\frac{5}{4}\right)\left(\frac{5}{3}\right) = 28. \end{aligned}$$

The normal gradient is $-\frac{1}{28}$, so the normal is $y + \frac{5}{4} = -\frac{1}{28}x$, or $x + 28y = -35$.

31)

$$\begin{aligned} y &= \sinh x \cosh x, \\ \frac{dy}{dx} &= \sinh^2 x + \cosh^2 x. \end{aligned}$$

Since $\sinh^2 x \geq 0$ and $\cosh^2 x \geq 1$ for all real x, $\frac{dy}{dx} \geq 1 > 0$, so the curve is always increasing.

32)

$$\begin{aligned} y &= (\cosh x - 3)\cosh x, \\ \frac{dy}{dx} &= (\cosh x - 3)\sinh x + \sinh x \cosh x \\ &= (2\cosh x - 3)\sinh x. \end{aligned}$$

Stationary points occur if $\cosh x = \frac{3}{2}$ or $\sinh x = 0$. For $\cosh x = \frac{3}{2}$,

$$x = \pm \ln\left(\frac{3}{2} + \sqrt{(\frac{3}{2})^2 - 1}\right) = \ln\left(\frac{3 \pm \sqrt{5}}{2}\right).$$

At either of these x-values, $y = (\frac{3}{2} - 3)\frac{3}{2} = -\frac{9}{4}$. Also, $\sinh x = 0$ gives $x = 0$ and $y = (1-3)\cdot 1 = -2$. The stationary points are

$$\left(\ln\frac{3+\sqrt{5}}{2}, -\frac{9}{4}\right), \quad \left(\ln\frac{3-\sqrt{5}}{2}, -\frac{9}{4}\right), \quad (0, -2).$$

33)

$$\begin{aligned} y &= \sinh\frac{x}{2}\left(2 - \sinh\frac{x}{3}\right), \\ y' &= -\frac{1}{3}\sinh\frac{x}{2}\cosh\frac{x}{3} + \frac{1}{2}\cosh\frac{x}{2}\left(2 - \sinh\frac{x}{3}\right), \\ y'' &= -\frac{13}{36}\sinh\frac{x}{2}\sinh\frac{x}{3} - \frac{1}{3}\cosh\frac{x}{2}\cosh\frac{x}{3} + \frac{1}{2}\sinh\frac{x}{2}. \end{aligned}$$

At $x = \ln 64$, this becomes

$$\begin{aligned} y'' &= -\frac{13}{36}\sinh(\ln 8)\sinh(\ln 4) - \frac{1}{3}\cosh(\ln 8)\cosh(\ln 4) + \frac{1}{2}\sinh(\ln 8) \\ &= -\frac{1365}{512} - \frac{1105}{384} + \frac{63}{32} = -\frac{5491}{1536} < 0. \end{aligned}$$

The curve is concave (concave downwards) at $x = \ln 64$.

34)

$$\begin{aligned} y &= (\sinh x - 6)\sinh x, \\ y' &= (\sinh x - 6)\cosh x + \cosh x \sinh x, \\ y'' &= 2\sinh^2 x + 2\cosh^2 x - 6\sinh x. \end{aligned}$$

Points of inflection satisfy

$$\begin{aligned} &2\sinh^2 x + 2\cosh^2 x - 6\sinh x = 0 \\ \Rightarrow\ &\sinh^2 x + \cosh^2 x - 3\sinh x = 0 \\ \Rightarrow\ &2\sinh^2 x - 3\sinh x + 1 = 0 \\ \Rightarrow\ &(2\sinh x - 1)(\sinh x - 1) = 0. \end{aligned}$$

Thus $\sinh x = \frac{1}{2}$ or $\sinh x = 1$, giving $x = \ln\left(\frac{1+\sqrt{5}}{2}\right)$ or $x = \ln(1+\sqrt{2})$. The corresponding y-coordinates are $-\frac{11}{4}$ and -5. The candidates are therefore $\left(\ln\frac{1+\sqrt{5}}{2}, -\frac{11}{4}\right)$ and $(\ln(1+\sqrt{2}), -5)$.

The two x-coordinates are approximately 0.481 and 0.881. The second derivative is positive at $x = 0$, negative at $x = 0.5$, and positive at $x = 1$, so its sign changes at both candidates. Both are points of inflection.

35)

$$\begin{aligned}\frac{d}{dx}(\tanh x) &= \frac{d}{dx}\left(\frac{\sinh x}{\cosh x}\right)\\ &= \frac{\cosh x\cosh x-\sinh x\sinh x}{\cosh^2 x}\\ &= \frac{\cosh^2 x-\sinh^2 x}{\cosh^2 x}=\frac{1}{\cosh^2 x}=\operatorname{sech}^2 x.\end{aligned}$$

36 a)

$$\begin{aligned}y &= \frac{\sinh x}{\cosh 2x},\\ \frac{dy}{dx} &= \frac{\cosh 2x\cosh x-\sinh x\cdot 2\sinh 2x}{\cosh^2 2x}\\ &= \frac{\cosh x\cosh 2x-2\sinh x\sinh 2x}{\cosh^2 2x}.\end{aligned}$$

36 b)

$$\begin{aligned}y &= \frac{\cosh x}{\sinh 2x},\\ \frac{dy}{dx} &= \frac{\sinh 2x\sinh x-\cosh x\cdot 2\cosh 2x}{\sinh^2 2x}\\ &= \frac{\sinh x\sinh 2x-2\cosh x\cosh 2x}{\sinh^2 2x}.\end{aligned}$$

36 c)

$y=\frac{x}{\sinh x}$, so $\dfrac{dy}{dx}=\dfrac{\sinh x-x\cosh x}{\sinh^2 x}$.

36 d)

$y=\frac{\cosh x}{x}$, so $\dfrac{dy}{dx}=\dfrac{x\sinh x-\cosh x}{x^2}$.

36 e)

$y=\frac{x+1}{\sinh 2x}$, so $\dfrac{dy}{dx}=\dfrac{\sinh 2x-2(x+1)\cosh 2x}{\sinh^2 2x}$.

36 f)

$y=\frac{\cosh 3x}{x^2+1}$, so $\dfrac{dy}{dx}=\dfrac{3(x^2+1)\sinh 3x-2x\cosh 3x}{(x^2+1)^2}$.

37)

$$y=\frac{\sinh 3x}{\sinh x},$$

$$\frac{dy}{dx} = \frac{3\sinh x\cosh 3x - \sinh 3x\cosh x}{\sinh^2 x}.$$

At $x = \ln 2$,

$$\begin{aligned}\left.\frac{dy}{dx}\right|_{x=\ln 2} &= \frac{3(\frac{3}{4})(\frac{65}{16}) - (\frac{63}{16})(\frac{5}{4})}{(\frac{3}{4})^2} \\ &= \frac{585/64 - 315/64}{9/16} = \frac{15}{2}.\end{aligned}$$

38)

$$\begin{aligned}y(0) &= \frac{0+\cosh 0}{1-\sinh 0} = 1, \\ \frac{dy}{dx} &= \frac{(1-\sinh 2x)(1+3\sinh 3x) - (x+\cosh 3x)(-2\cosh 2x)}{(1-\sinh 2x)^2}, \\ \left.\frac{dy}{dx}\right|_{x=0} &= 3.\end{aligned}$$

The tangent is $y - 1 = 3(x - 0)$, so $y = 3x + 1$.

39)

$$\begin{aligned}y(0) &= \frac{\tanh 0 + 6}{\cosh 0 + 3} = \frac{3}{2}, \\ \left.\frac{dy}{dx}\right|_{x=0} &= \frac{3(\cosh 0 + 3)\operatorname{sech}^2 0 - 2(\tanh 0 + 6)\sinh 0}{(\cosh 0 + 3)^2} = \frac{3}{4}.\end{aligned}$$

The normal gradient is $-\frac{4}{3}$, so $y - \frac{3}{2} = -\frac{4}{3}(x - 0)$ and $8x + 6y = 9$.

40 a)

$$\begin{aligned}f(x) &= \frac{x+1}{\sinh x}, \\ f'(x) &= \frac{\sinh x - (x+1)\cosh x}{\sinh^2 x}.\end{aligned}$$

Stationary points satisfy $\sinh x - (x + 1)\cosh x = 0$. Dividing by $\cosh x$ gives $\tanh x = x + 1$.

40 b)

There is one point of intersection between $y = \tanh x$ and $y = x + 1$, so the curve $y = f(x)$ has only one stationary point.

41)

$$\begin{aligned}y &= \frac{\sinh x + 1}{\tanh x}, \\ y' &= \frac{\tanh x\cosh x - (\sinh x + 1)\operatorname{sech}^2 x}{\tanh^2 x}\end{aligned}$$

$$= \frac{\sinh x - (\sinh x + 1)/\cosh^2 x}{\tanh^2 x}.$$

At a stationary point,

$$\begin{aligned} \sinh x \cosh^2 x - \sinh x - 1 &= 0 \\ \Rightarrow \sinh x(\sinh^2 x + 1) - \sinh x - 1 &= 0 \\ \Rightarrow \sinh^3 x &= 1 \\ \Rightarrow \sinh x = 1 \Rightarrow x &= \ln(1+\sqrt{2}). \end{aligned}$$

$$\begin{aligned} y &= \frac{\sinh(\ln(1+\sqrt{2})) + 1}{\tanh(\ln(1+\sqrt{2}))} \\ &= \frac{\frac{1+\sqrt{2}-(-1+\sqrt{2})}{2} + 1}{\frac{1+\sqrt{2}-(-1+\sqrt{2})}{1+\sqrt{2}+(-1+\sqrt{2})}} = 2\sqrt{2}. \end{aligned}$$

The stationary point is $(\ln(1+\sqrt{2}), 2\sqrt{2})$.

42)

$$\begin{aligned} y &= \frac{\cosh x}{4 - \sinh x}, \\ y' &= \frac{(4-\sinh x)\sinh x + \cosh^2 x}{(4-\sinh x)^2} \\ &= \frac{4\sinh x + 1}{(4-\sinh x)^2}. \end{aligned}$$

Set $y' = \frac{1}{16}$ and let $s = \sinh x$:

$$\begin{aligned} 16(4s+1) &= (4-s)^2 \\ 64s + 16 &= 16 - 8s + s^2 \\ 0 &= s^2 - 72s = s(s-72). \end{aligned}$$

Thus $\sinh x = 0$ or $\sinh x = 72$. The first gives the stated point $x = 0$; the other gives $x = \ln(72 + \sqrt{72^2+1}) = \ln(72 + \sqrt{5185})$.

43)

$$\begin{aligned} y &= \frac{4}{\cosh x}, \\ y' &= -\frac{4\sinh x}{\cosh^2 x}, \\ y'' &= \frac{8\sinh^2 x - 4\cosh^2 x}{\cosh^3 x}. \end{aligned}$$

Points of inflection satisfy $8\sinh^2 x - 4\cosh^2 x = 0$, so $2\tanh^2 x - 1 = 0$ and $\tanh x = \pm\frac{1}{\sqrt{2}}$. This gives $x = \ln(1+\sqrt{2})$ or $x = \ln(-1+\sqrt{2})$. At both points, $y = \frac{4}{\cosh x} = 2\sqrt{2}$. The second derivative changes sign at each point, so they are both points of inflection:

$$(\ln(1+\sqrt{2}), 2\sqrt{2}), \quad (\ln(-1+\sqrt{2}), 2\sqrt{2}).$$

44)

$$\begin{aligned}\frac{d}{dx}(\coth x) &= \frac{d}{dx}\left(\frac{\cosh x}{\sinh x}\right)\\ &= \frac{\sinh^2 x - \cosh^2 x}{\sinh^2 x} = -\frac{1}{\sinh^2 x} = -\operatorname{cosech}^2 x.\end{aligned}$$

45)

$$\begin{aligned}\frac{d}{dx}(\operatorname{cosech} x) &= \frac{d}{dx}\left(\frac{1}{\sinh x}\right)\\ &= -\frac{\cosh x}{\sinh^2 x} = -\coth x \operatorname{cosech} x.\end{aligned}$$

46)

$$\begin{aligned}\frac{d}{dx}(\operatorname{sech} x) &= \frac{d}{dx}\left(\frac{1}{\cosh x}\right)\\ &= -\frac{\sinh x}{\cosh^2 x} = -\tanh x \operatorname{sech} x.\end{aligned}$$

47)

$$\begin{aligned}y &= 2x + \operatorname{sech} x,\\ y' &= 2 - \tanh x \operatorname{sech} x,\\ y'' &= \tanh^2 x \operatorname{sech} x - \operatorname{sech}^3 x\\ &= \operatorname{sech} x(\tanh^2 x - \operatorname{sech}^2 x).\end{aligned}$$

Since $\operatorname{sech} x \neq 0$, points of inflection satisfy $\tanh^2 x - \operatorname{sech}^2 x = 0$, hence $\sinh^2 x - 1 = 0$, so $\sinh x = \pm 1$. Therefore $x = \ln(1+\sqrt{2})$ or $x = \ln(-1+\sqrt{2})$. The second derivative changes sign on either side of both values, so there are two points of inflection.

48)

$$\begin{aligned}\frac{d}{dx}\left(\ln \tanh \frac{x}{2}\right) &= \frac{\frac{1}{2}\operatorname{sech}^2 \frac{x}{2}}{\tanh \frac{x}{2}}\\ &= \frac{1 - \tanh^2 \frac{x}{2}}{2\tanh \frac{x}{2}}\\ &= \frac{\cosh^2 \frac{x}{2} - \sinh^2 \frac{x}{2}}{2\sinh \frac{x}{2}\cosh \frac{x}{2}}\\ &= \frac{1}{\sinh x} = \operatorname{cosech} x.\end{aligned}$$

49)

$$y = \arctan(4e^{5x}),$$
$$\frac{dy}{dx} = 20e^{5x}\frac{1}{(4e^{5x})^2+1} = \frac{20}{16e^{5x}+e^{-5x}}.$$

$$p\sinh 5x + q\cosh 5x \equiv \frac{1}{2}(p+q)e^{5x} + \frac{1}{2}(-p+q)e^{-5x}.$$

Equating coefficients gives $\frac{1}{2}(p+q) = 16$, so $p+q = 32$, and $\frac{1}{2}(-p+q) = 1$, so $-p+q = 2$. Solving gives $p = 15$, $q = 17$. Therefore,

$$\frac{dy}{dx} = \frac{20}{15\sinh 5x + 17\cosh 5x}.$$